

## DS 5030: Lab 1 Report: Genre Retrieval System

When I evaluated the precision@3 of raw dot product, cosine similarity, and tf-idf cosine similarity, I was surprised by how low the results were, especially for tf-idf. Only about 21% of tf-idf cosine similarity recommendations matched the correct genre, and that percentage only jumped to 23% when looking at just the #1 matches. This suggests that word count patterns and word uniqueness aren't strong enough predictors of genre, so we need to find another variable that can aid this.

I looked at the original *songs* data frame to find a column that might help enhance our search for similar genre and ended up going with *top\_tags* because they appeared to be user-written and thus would provide more subjective data for each song:

```
genre
blues      [blues, classic rock, rock, hard rock, blues r...
classical  [Baladas, Classical, christian rock, christian...
country    [Martina McBride, country, female vocalists, c...
electronic [dance, techno, party, electronic, synth pop, ...
folk       [folk, female vocalists, indie, Mellow, love s...
hip hop    [underground hip-hop, hip hop, Hip-Hop, belly ...
jazz       [jazz, easy listening, pop, jamie cullum, jazz...
latin      [latin, pop, latin pop, Pop/Rock, mexican, mar...
metal      [metal, Nu Metal, alternative metal, hardcore,...
pop        [60s, oldies, pop, female vocalists, rock n ro...
punk       [punk, covers, Punk Favorites, cover favorites...
r&b/soul   [soul, motown, 70s, marvin gaye, funk, soul, f...
reggae     [ska, ska punk, punk, Mellow, reggae, reggae, ...
rock       [rock iberico, mpb, brasil, classic rock, expe...
Name: top_tags, dtype: object
```

*Figure 1. Initial exploration of top\_tags by genre*

Specifically, some tags had combined genre descriptions like “blues rock” or “Latin pop,” which would add nuance to a system that already uses tf-idf cosine similarity to match for word patterns/uniqueness. **Importantly, some of the user tags used the exact genre name as a tag, which seemed a little bit like cheating since genre is our ground truth for a recommendation.** Because of this, I removed any tags from a song that had an exact match for its same genre name but kept interesting ones like “blues rock” or “underground hip-hop” because sharing these niche-r terms would probably be rare and thus a potentially high-value similarity identifier. Songs were allowed to maintain basic genre tags that did not match their own (**e.g., songs in the “rock” genre can have a “blues” tag or a “blues rock” tag, but not a “rock” tag**).

To implement this idea for my genre retrieval system, I consulted ChatGPT on the easiest way to standardize the *top\_tag* column and even helped correct its code to ensure that “r&b/soul” genre tags were properly filtered out. ChatGPT also provided a recommendation to use Jaccard

similarity as a measure of comparing how tags overlap between songs (it compares two sets by dividing the number of shared elements by the total number of unique elements, giving a score between 0-1). I then researched Jaccard similarity on my own to ensure the method made sense, and I investigated the spread of *top\_tag* sizes to confirm that there were no crazy outliers (all *top\_tag* columns had 0-5 tags per song, which was helpful). ChatGPT also helped convert my previous tf-idf similarity code to add a Jaccard similarity calculation (and make it more efficient than my previous attempts), and I also collaborated with it to include a way to change the weights, so that I could experiment with tf-idf cosine similarity vs Jaccard similarity holding more importance.

I first tested my new system as 0.8(tf-idf cosine) and 0.2(tags), and the precision@3 jumped to 33%, already improving on baseline. I then tested different weights before landing on 0.3(tf-idf cosine) and 0.7(tags) as the best combination, which surprised me, as I anticipated that tf-idf cosine similarity might be the most meaningful of the two.

```
TF-IDF cosine baseline: 0.211
80/20: 0.333
70/30: 0.451
60/40: 0.529
50/50: 0.595
40/60: 0.607
30/70: 0.615
20/80: 0.608
10/90: 0.602
Jaccard tag baseline: 0.573
```

*Figure 2. Precision@3 results for different weights*

At my system's best genre-matching weights (0.3/0.7), over 61% of recommendations were from the matching genre. Since genre is the ground truth for this assignment, I was able to triple the baseline output of tf-idf cosine similarity. I was very skeptical of my results, so I asked Chat-GPT for simple ways to cover my bases, and I was able to confirm that no song had its same genre tag and that all code to exclude songs of the same artist was correct. To reiterate, even though I removed exact genre tag matches from songs (a song in genre "rock" can't have the "rock" tag), I kept tags that included more stylistic genre names like "blues rock" or "country pop" because two songs sharing these more specific tags would be meaningful without fully pointing to a genre. In fact, because I've removed exact genre-matches, these specific, descriptive terms seem like they would be harder to find exact matches for due to their stylistic emphasis.

Here's a breakdown of genre matches by genre for each similarity measure:

|            | dot_hit  | cos_hit  | tfidf_cos_hit | tag_tfidf_hits_30_70 |            | dot_hit  | cos_hit  | tfidf_cos_hit | tag_tfidf_00_1_hits |
|------------|----------|----------|---------------|----------------------|------------|----------|----------|---------------|---------------------|
| genre      |          |          |               |                      | genre      |          |          |               |                     |
| hip hop    | 0.525000 | 0.458333 | 0.566667      | 0.850000             | metal      | 0.008333 | 0.191667 | 0.150000      | 0.808333            |
| latin      | 0.216667 | 0.666667 | 0.575000      | 0.841667             | hip hop    | 0.525000 | 0.458333 | 0.566667      | 0.783333            |
| metal      | 0.008333 | 0.191667 | 0.150000      | 0.800000             | punk       | 0.008333 | 0.041667 | 0.116667      | 0.783333            |
| punk       | 0.008333 | 0.041667 | 0.116667      | 0.783333             | rock       | 0.000000 | 0.075000 | 0.108333      | 0.675000            |
| rock       | 0.000000 | 0.075000 | 0.108333      | 0.675000             | blues      | 0.008333 | 0.141667 | 0.183333      | 0.658333            |
| blues      | 0.008333 | 0.141667 | 0.183333      | 0.666667             | latin      | 0.216667 | 0.666667 | 0.575000      | 0.625000            |
| folk       | 0.000000 | 0.108333 | 0.083333      | 0.608333             | jazz       | 0.000000 | 0.116667 | 0.133333      | 0.608333            |
| jazz       | 0.000000 | 0.116667 | 0.133333      | 0.583333             | folk       | 0.000000 | 0.108333 | 0.083333      | 0.541667            |
| country    | 0.000000 | 0.100000 | 0.166667      | 0.550000             | country    | 0.000000 | 0.100000 | 0.166667      | 0.483333            |
| r&b/soul   | 0.450000 | 0.250000 | 0.275000      | 0.500000             | electronic | 0.050000 | 0.133333 | 0.116667      | 0.466667            |
| classical  | 0.000000 | 0.233333 | 0.250000      | 0.475000             | r&b/soul   | 0.450000 | 0.250000 | 0.275000      | 0.458333            |
| electronic | 0.050000 | 0.133333 | 0.116667      | 0.466667             | reggae     | 0.016667 | 0.041667 | 0.108333      | 0.400000            |
| pop        | 0.016667 | 0.116667 | 0.116667      | 0.408333             | classical  | 0.000000 | 0.233333 | 0.250000      | 0.383333            |
| reggae     | 0.016667 | 0.041667 | 0.108333      | 0.400000             | pop        | 0.016667 | 0.116667 | 0.116667      | 0.358333            |

Figure 3. Percentages of giving a same-genre match by genre (for .3 and .7 weights, as well as Jaccard tag baseline)

As shown above, the combination tf-idf cosine(0.3) and Jaccard(0.7) method managed at least 40% for genre matches, with hip hop, latin, and metal genres having over 80% accuracy. I've also attached the Jaccard similarity baseline results, which only beat out the joint genre recommendation method for the metal genre (0.808 vs. 0.800), but *barely*.

There are some limitations to my approach, including lack of context about the *top\_tags* column (e.g., Where did these tags come from? How many users contributed? How were tags narrowed down to five?) and potential nuances lost when standardizing the tag column. This approach is very dependent on the *top\_tag* column, so replicating this on a different data set may produce wildly different results (and some data sets may not have a comparable tag column at all). Additionally, my investigation into top tags by genre was not the most extensive, so there could be details there that skewed the results. Nonetheless, using these user-defined tags was interesting to me and proved to be a fairly successful endeavor in providing genre-matching recommendations. I even wonder if the specificity of some of these tags lends themselves better to a broader question of recommendation that doesn't use genre as a baseline. For example, some rock songs and blues songs used the tag "blues rock", despite technically being labeled as different genres. This suggests that a recommendation system should not only consider genre as the ground truth threshold because it denies cross-genre similarities and preferences noted by listeners/users. For the purpose of this assignment, however, I understand the need to establish truth somewhere.